



The entire setup. Don't get awed by the low temperature, though—this shot was taken without the acrylic sheet

We found that the ATI Radeon X1300XL graphics card we used had active cooling, but due to its design, sent its exhaust (hot air) towards the centre of the cabinet. When we compared it to the newer design of the X1900XT, which occupies two slots but sends its exhaust directly out the back of the cabinet, we found that older graphics cards really leave a lot to be desired in terms of airflow!

We also noticed that there was hardly any air flowing over or under the hard drives or the DVD-Writer. Now a DVD-Writer may not need cooling, but hard drives certainly do!

Fanning The Front

We found that most cabinets have a space to attach a front cabinet fan. However, the front is blocked by the Power and Restart buttons, along with the hard drive and power LEDs. Front USB ports also block the front of most cabinets.

There are big gaps in the front and along the sides of cabinets, which were made obvious by the smoke billowing out of these gaps when the PC was switched off. We decided that adding a front fan was the next step, and these gaps should allow the fan to draw enough air in. The first hurdle was the fact that the location of the front fan was obstructing the hard drive placement. We couldn't push the hard drives all the way in, and settled for them sticking out a little towards the motherboard in favour of the additional fan.

It worked! Soon after the cabinet was filled with smoke, we found that switching it on made the smoke accumulated around the hard drive disappear almost instantly and flow towards the rear cabinet fan. If anything, the flow of air from the X1300XL was the only obstruction to smooth airflow. When we removed the X1300XL and used the onboard graphics, we found that air flowed much better! We also saw that smoke from outside the front of the cabinet was being drawn in through the gaps in the front, so despite being covered, the front fan was working well.

Cooler Disks

The next step was to cool the hard drives. We attached the hard drive coolers we'd bought, and immediately found that they refused to sit comfortably in their bays. The problem was that even if we managed to muscle them in, because of the extra height the hard drives

Tip

Buying A Cabinet

When buying a cabinet, try not to choose a small form factor cabinet, especially if you want powerful hardware. The space savings are not worth running your components hot. Also, a lot of cabinets come with top fans already attached. Some even come with front fans. If you don't want to have to fiddle about inside the cabinet, make sure you get yourself a cabinet with at least one top and one front fan, in addition to a large rear fan.